

BIG IDEAS
FOR EVERY SPACE

From 0 to PROFINET IRT in no TIME

TPS-1 : A SINGLE CHIP INTERFACE FOR PROFINET IO



Renesas Tiger PROFINET Single-chip (TPS1) device enables the move from PROFIBUS to the new Ethernet based communication standards like PROFINET in no time.

It minimizes cost efforts for implementing and provides device manufacturers with a fast, easy and cost-saving integration of a PROFINET IO device interface. TPS1 device comes with an integrating CPU, a 2-port switch supporting latest PROFINET specifications, the Ethernet PHYs and peripheral modules to offer best-in-class IRT solution.



Benefits

- Minimal licensing cost & technical support
- Minimal investment in Test Equipment
- Minimal software development effort
- Cost saving on integration – tools included
- Faster implementation & development – Complete communication system solution

Specifications of TPS-1 System on Module from Calixto

- Renesas TPS-1 & RX631 (32 Bit, 100MHz)
- 1/2/4 MB NOR Flash for TPS-1
- 2 x PROFINET ports
- 42 PROFINET IO (TPS-1)
- Up to 3 UART, 1 USB Device, 1 CAN, 4 PWM
- 1 x I²C & 3 x 12-bit ADC (RX631)
- JTAG (TPS-1) & FINE interface (RX631)
- On board RTC Chip
- 3.3V operation
- Dimension 40x38mm
- 0.5 mm pitch 2 x 80 pin connectors for interfacing to Carrier board



Applications

- Industrial Automation
- Electrical Automation
- Power Meters
- Compact & Modular Remote IO



Features

- Integrated PROFINET CPU
- Compliant with Conformance Class C
- 2 external ports, 100 Mbit/s, full duplex
- 2 integrated PHYs with auto negotiation, auto crossover
- Integrated IRT switch, 8 priority levels
- Supports RJ45 or fiber optic interfaces
- Fiber optic diagnosis via I²C interface per port
- IRT bridge-delay < 3 μ s
- Hardware support for PROFINET protocols including PTCP and LLDP
- Versatile host interface for serial or parallel connection of external CPUs or local inputs/outputs
- Small package (15 x 15 mm), 1 mm ball pitch